
Cover Letter

5 September 2026

Dear Editors,

Please consider our Article, "A prospective assay reveals scaffold-driven hypothesis-space expansion," for publication in *Nature Machine Intelligence*.

When an AI-assisted discovery system produces a successful explanation, the result alone does not identify whether its decisive content came from the language model, programmed search or supplied domain knowledge. We introduce a prospective, replayable assay that separates validation of a representational change from attribution of the executable content used to evaluate it.

The assay fixes an incumbent representation language and an observation-optimal comparator. A candidate must fall outside the language's structural rules, fit the observations and outperform the comparator on an intervention committed before its outcome is revealed and on separate held-out tests. This links representational change to verifiable predictive consequences rather than judgments of creative wording.

In controlled synthetic worlds, deterministic compositional search paired with a fixed, family-aligned motif-to-basis realizer produced validated candidates in all 400 known-family worlds and 100 worlds from one search-side holdout. A post-hoc code-path audit and inference-free replay reproduced every candidate verdict from this search condition without model output: the scaffold supplied the representations, fitted expressions and interventions. Interface auditing exposed the complementary attribution problem. Legacy model-proposal failures occurred before executable evaluation; a redesigned, grammar-constrained interface yielded validated proposals in 15 of 96 worlds, compared with 16 of 96 for random composition on the same panel. These findings distinguish interface failures from evaluated proposals, as well as system performance from model contribution.

The manuscript organizes these findings into a reusable evidence-recording procedure: commit the evaluated theory and predictions before outcome reveal, trace model- and scaffold-authored fields, replay specified component interventions, and inventory supplied operators and realization knowledge. This separates system-level escape, model contribution and representation invention, and makes both successful outputs and interface-limited failures inspectable.

The demonstration is restricted to closed, noiseless procedural worlds with supplied family-aligned realization knowledge and one adjacent search-side holdout; application to independently authored systems remains an external-validation question. Code, evaluation records and replay materials are available through the public research release described in the manuscript. We believe this combination of prospective evaluation and executable-content attribution is relevant to *Nature Machine Intelligence* readers working on AI evaluation, symbolic reasoning and agentic discovery because it makes explicit the evidence required to assign credit within model-scaffold systems.

This manuscript has not been published in a journal and is not under consideration by another journal. The authors declare no competing interests, report no specific funding and have approved the manuscript, its submission and the disclosed use of AI assistance.

Sincerely,

Jing-Rung Huang

Corresponding author, on behalf of both authors

Department of Computer Science and Information Engineering

National Cheng Kung University, Tainan 701, Taiwan

p78084063@mail.ncku.edu.tw

ORCID: 0000-0003-4776-3550